HIGH PRESSURE TURBINE ROTOR STAGE 1 DISK

TEST FOR CRACKS AND OTHER DEFECTS WITH ULTRASONIC

Applicability

Engine type(s): LEAP-1B
Engine model(s): ALL

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1. REFERENCES

Table: References

Technical Publication
LEAP-1B-72-52-40-01A-25BA-C
LEAP-00-70-35-13-01A-355A-D
LEAP-1B-72-52-40-01A-664B-C

2. REQUIRED CONDITIONS

Table: Required conditions

Action / Condition	Reference
Clean the stage 1 disk and remove the corrosion resistant coating.	LEAP-1B-72-52-40-01A-25BA-C

3. SUPPORT EQUIPMENT

Table: Support equipment

Name	Tool Number	Reference	Quantity
Standard Tools			
Ultrasonic Immersion Inspection System, Refer to: LEAP-00-70-35-13-01A-355A-D	--	Not Applicable	AR
Ultrasonic Transducer TYPE A Olympus/Harisonic	I3-1006-T, TLC IS1010GA, or UTX IX-139	Not Applicable	1
Ultrasonic Transducer TYPE B Olympus/Harisonic	I7-0512-T, TLC IX0519GB, or UTX IX-141	Not Applicable	1
Ultrasonic Mirror and Adaptor 1.50 in (38.1 mm) or 2.00 in (50.8 mm) for TYPE A Transducer	--	Not Applicable	AR
Ultrasonic Mirror and Adaptor 3.50 in (88.9 mm) or 5.0 in (127 mm) for TYPE B Transducer	--	Not Applicable	AR
Calibration Block SGP-044 SDH (Side Drilled Hole)	--	Not Applicable	AR
Calibration Block 4013316-417 FBH (0.060 in (1.52 mm) - 1.50 in (38.1 mm) Flat Bottom Holes)	--	Not Applicable	AR
Part Fixtures (Dependent on UT Inspection System)	--	Not Applicable	AR

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4. CONSUMABLES, MATERIALS AND EXPENDABLES

No supplies

5. SPARES

No Spares

Safety conditions

NOTE: *Inspection facilities are responsible for part fixturing and approved to manufacture their own fixtures. Part fixtures and part risers have been designed to work with the Nipscan System. These items can be made to adapt to other approved UT inspection systems. For additional information regarding fixture, please contact the E-mail that follows:*
E-mail: ndttooling@ge.com

6. PROCEDURE

A. General

(1) This procedure gives instructions to do a component level immersion ultrasonic inspection of the stage 1 **High Pressure Turbine *(HPT)*** rotor disk (stage 1 disk) (01-250, 72-52-00) (**SIN:** 158A0).

(2) To get tooling, additional items, or replacement items, contact the following:

E-mail: ndttooling@ge.com

(3) Changes to this procedure or use of alternative equipment can result in a loss of detection of cracks. Alternative equipment and procedures must be evaluated and approved by CFM.

NOTE: *You can use equivalent alternatives for standard tools, equipment, and consumable materials.*

(4) Do a pre-inspection part preparation/cleaning. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(5) Make sure that the inspection system agrees with all the necessary mechanical accuracies. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

B. Equipment calibration

(1) Part material is Rene 88, for this material it is necessary to apply a correction factor for calibration. Use the Inco 718 correction factor.

(2) For longitudinal calibration do as follows:

(a) Install the ultrasonic transducer TYPE A for the longitudinal wave calibration.

(b) Calibrate the longitudinal scans. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(c) Record all the calibration data on the applicable calibration form. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(3) For shear wave calibration do as follows:

(a) Install the ultrasonic transducer TYPE B for the shear wave calibration.

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(b) Calibrate the 45 and 65 degree scans. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(c) Record all the calibration data on the applicable calibration form. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

C. Immersion ultrasonic inspection

(1) For longitudinal inspection (ultrasonic transducer TYPE A) do as follows:

(a) Follow the inspection routines. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(b) Use the applicable index rate, scan speed, and gain identified for longitudinal inspection. Refer to **Figure 1**.

(c) Use the ultrasonic transducer TYPE A gate settings. Refer to **Figure 3**.

(d) If necessary, install the ultrasonic mirror and adaptor 1.50 in **(**38.1 mm**)** or 2.00 in **(**50.8 mm**)** such that the sound beam is normal to the inspection surface.

(e) It will be necessary to adjust the back of the gate to 1.50 in **(**38.1 mm**)** or to the back wall, if it is closer while the inspection is in process. The back-wall gate must not be more than 1.50 in **(**38.1 mm**)** while you scan.

(f) Scan the hardware in accordance to **Figure 6** and save all the C-scan data for each scan.

(g) Do a post calibration check. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(h) Record the inspection results in the Form 1 of this procedure. Refer to **Figure 7**.

(2) For shear wave inspection (ultrasonic transducer TYPE B) do as follows:

(a) Follow the inspection routines. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(b) Refer to **Figure 2** for shear scan offsets.

(c) Use the applicable index rate, scan speed, and gain. Refer to **Figure 1**.

(d) Use the ultrasonic transducer TYPE B gate settings. Refer to **Figure 3**.

(e) For the bore scans (Zone C) do as follows:

1 Install the ultrasonic mirror and adaptor 3.50 in **(**88.9 mm**)** or 5.0 in **(**127 mm**)** such that the sound beam is normal to the inspection surface.

2 For 45-degree shear, it will be necessary to adjust the back of the gate to 2.00 in **(**50.8 mm**)** or to the back wall, if it is near while the inspection is in process.

3 For 45-degree shear, the back-wall gate must not be more than 2.00 in **(**50.8 mm**)** while you scan.

(f) For the transition areas and web scans (Zones A, B, D, and E) do as follows:

1 Remove the mirror and scan the areas in accordance with the scan plans. Refer to **Figure 6**.

2 For 45-degree shear, it will be necessary to adjust the back of the gate to 2.00 in **(**50.8 mm**)** or to the back wall, if it is near while the inspection is in process.

3 For 45-degree shear, the back-wall gate must not be more than 2.00 in **(**50.8 mm**)** while you scan.

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(g) Scan the hardware in accordance with **Figure 6** and save all the C-scan data for each scan.

(h) Do a post calibration check. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(i) Record the inspection results in the Form 1 of this procedure. Refer to **Figure 7**.

D. Evaluation of the ultrasonic indications

(1) All indications must be evaluated at the gain levels specified in **Figure 4**.

(a) Gain for longitudinal scan must be at calibration gain.

(b) Gain levels for 45 and 65 degree shear scans must be at calibration gain +6 dB.

(2) Evaluate all the repeatable discrete indications which are equal to or more than 32 percent **Full Screen Height *(FSH)*** . Refer to **Figure 5**.

(a) Peak the response by translation and rotation only.

(b) Reject all the hardware for discrete indications which are repeatable and equal to or more than 40 percent ***FSH***.

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MODE	INSPECTION GAIN	MAX INDEX RATE	MAX SCAN SPEED	ZONES
LONGITUDINAL	CALIBRATION	0.020 IN./REV	18 IN./SEC	**C**
45-DEGREE SHEAR	CALIBRATION +6dB	0.040 IN./REV	18 IN./SEC	**A**, **B**, **C**, **D**, AND **E**
65-DEGREE SHEAR	CALIBRATION +6dB	0.040 IN./REV	18 IN./SEC	**B**, **C**, **D**, AND **E**

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Inspection parameters
Figure 1

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	0 LONGITUDINAL	45-DEGREE SHEAR	65-DEGREE SHEAR
NON-MIRROR	0 DEGREES	20 DEGREES	26.2 DEGREES
Y MOVEMENT BORE FROM CENTER (MIRROR AT 0 DEGREES)	0 IN.	0.76 IN. (19.3 mm)	0.98 IN. (24.9 mm)

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Offsets
Figure 2

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TYPE A TRANSDUCER		
MODE	GATE	ZONE C
LONG	START	0.100
	END	1.500

TYPE B TRANSDUCER						
MODE	GATE	ZONE A	ZONE B	ZONE C	ZONE D	ZONE E
45-DEGREE SHEAR	START	0.100	0.100	0.100	0.100	0.100
	END	2.000 OR BW	2.000 OR BW	2.000 OR BW	2.000 OR BW	2.000 OR BW
65-DEGREE SHEAR	START	N/A	0.040	0.040	0.040	0.040
	END	N/A	0.450	0.450	0.450	0.450

NOTE:
GATE START SETTINGS ARE CONTINGENT ON FRONT SURFACE BASELINE NOISE. DO THE BEST EFFORT TO PUT AS CLOSE TO FRONT SURFACE AS POSSIBLE. DOCUMENT ANY GATE SETTING ADJUSTMENTS ON FORM 1 INSPECTION WORKSHEET.

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Gate settings
Figure 3

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MODE	EVALUATION GAIN
LONGITUDINAL	CALIBRATION GAIN
45 DEGREE	CALIBRATION GAIN +6dB
65 DEGREE	CALIBRATION GAIN +6dB

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Gain levels for evaluation
Figure 4

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INDICATION EVAL LIMITS (FSH)	REJECT LIMIT (FSH)
32 PERCENT OR MORE	40 PERCENT

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Reject limits
Figure 5

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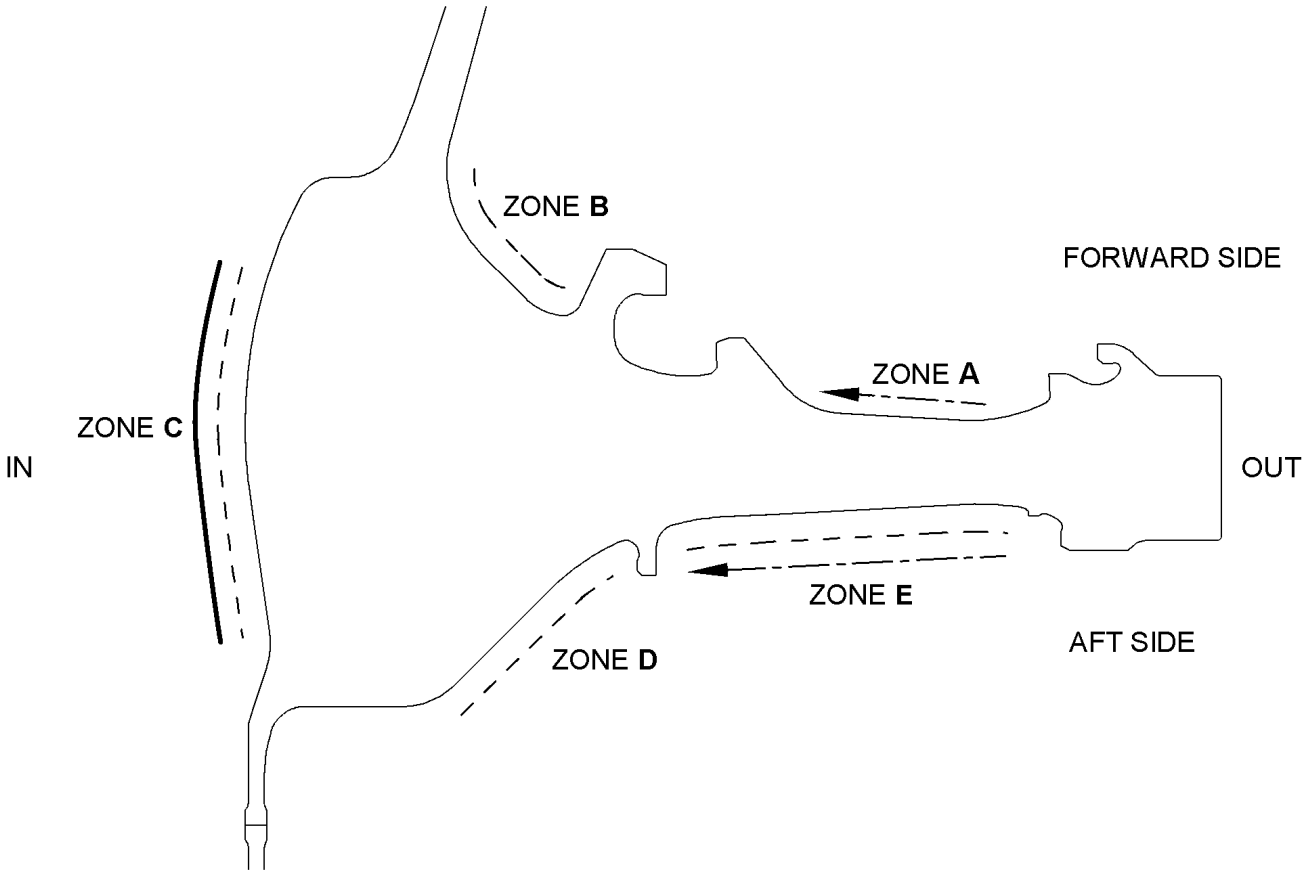
P/N: 2600M21G02



ZONES	INSPECTION AREA
ZONE **A**, **B**, **D**, AND **E**	WEB AND TRANSITION AREA
ZONE **C**	BORE

INSPECTION MODE	
CIRCUMFERENTIAL SHEAR (CLOCKWISE (CW) AND COUNTERCLOCKWISE (CCW))	- - - - - - - - - - -
RADIAL SHEAR (IN AND/OR OUT)	—— - —— - —— - —— - ——
LONGITUDINAL	————————

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Stage 1 disk - Immersion ultrasonic inspection plan
Figure 6

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FORM 1. HPT STAGE 1 DISK INSPECTION WORKSHEET

DATE:		OPERATOR:	
SITE:		INST TYPE:	
HARDWARE P/N:		HARDWARE S/N:	
SDH CAL STD P/N:		TYPE **A** TRANS P/N:	
SDH CAL STD S/N:		TYPE **A** TRANS S/N:	
FBH CAL STD P/N:		TYPE **B** TRANS P/N:	
FBH CAL STD S/N:		TYPE **B** TRANS S/N:	

TYPE **A** INSPECTION (LONGITUDINAL)

ZONE	ACCEPT	REJECT
C		

TYPE **B** INSPECTION (SHEAR)

ZONE	45-DEGREE CW		45-DEGREE CCW		65-DEGREE CW		65-DEGREE CCW		45-DEGREE RADIAL IN		45-DEGREE RADIAL OUT	
	ACCEPT	REJECT	ACCEPT	REJECT	ACCEPT	REJECT	ACCEPT	REJECT	ACCEPT	REJECT	ACCEPT	REJECT
A	X	X	X	X	X	X	X	X			X	X
B									X	X	X	X
C									X	X	X	X
D									X	X	X	X
E											X	X

NOTES:

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Form 1
Figure 7

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7. REQUIREMENTS AFTER JOB COMPLETION

Table: Requirements after job completion

Action / Condition	Reference
If all inspections are complete, apply the corrosion resistant coating again.	LEAP-1B-72-52-40-01A-664B-C

End of data module

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